

Singular Value Decomposition

Definition, Suppose $T \in L(V)$. The singular values of T are the (nonnegative) square roots of the eigenvalues of T^*T , with each eigenvalue λ repeated $\dim E(\lambda, T^*T)$ times. That is if

$$T^*T r_j = \lambda_j r_j, \quad j \in [1:n]$$

$$\lambda_j \geq 0.$$

$\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}$ are singular values.

Exercise. Show that the singular values of T are the eigenvalues of $\sqrt{T^*T}$ with each eigenvalue λ repeated $\dim E(\lambda, \sqrt{T^*T})$ times.

Example. Define $T \in L(F^4)$ by

$$T(z_1, z_2, z_3, z_4) = (0, 3z_1, 2z_2, -3z_4).$$

$$\begin{aligned} \Rightarrow \langle z, T^*w \rangle &= \langle Tz, w \rangle \\ &= 3z_1 \bar{w}_2 + 2z_2 \bar{w}_3 - 3z_4 \bar{w}_4 \\ &= \langle z, (3w_2, 2w_3, 0, -3w_4) \rangle \end{aligned}$$

$$\Rightarrow T^*(w) = (3w_2, 2w_3, 0, -3w_4)$$

$$\begin{aligned} T^*T(z) &= T^*(0, 3z_1, 2z_2, -3z_4) \\ &= (9z_1, 4z_2, 0, -9z_4). \end{aligned}$$

For the standard basis $B = \{e_1, e_2, e_3, e_4\}$,

$$[T^*T]_B = \begin{bmatrix} 9 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}. \quad \begin{aligned} E(9, T^*T) &= \text{span}(e_1, e_4) \\ E(4, T^*T) &= \text{span}(e_2) \\ E(0, T^*T) &= \text{span}(e_3) \end{aligned}$$

So the singular values of T are $3, 2, 0$.

Theorem. (Singular value Decomposition)

Suppose $T \in L(V)$ has strictly positive singular values $s_1, \dots, s_n$. Then there exist orthonormal bases $e_1, \dots, e_n$ and $f_1, \dots, f_n$ of V such that

$$Te_j = s_j f_j, \quad j \in [1:n]$$

or equivalently

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_n \langle v, e_n \rangle f_n,$$

for every $v \in V$.

Proof. $n = \dim V$. Because T^*T is a positive operator, the spectral theorem implies that there exists an orthonormal basis $e_1, \dots, e_n$ of V with

$$T^*T e_k = \lambda_k e_k, \quad k \in [1:n], \quad s_k = \sqrt{\lambda_k}$$

For each $k \in [1:n]$ define

$$f_k = \frac{Te_k}{s_k}.$$

If $j, k \in [1:n]$ then

$$\begin{aligned} \langle f_j, f_k \rangle &= \frac{1}{s_j s_k} \langle T e_j, T e_k \rangle = \frac{1}{s_j s_k} \langle T^* T e_j, e_k \rangle \\ &= \frac{1}{s_j s_k} \langle \lambda_j e_j, e_k \rangle = \frac{s_j}{s_k} \langle e_j, e_k \rangle \\ &= \begin{cases} 0 & \text{if } j \neq k \\ 1 & \text{if } j = k. \end{cases} \end{aligned}$$

Thus $f_1, \dots, f_n$ is an orthonormal basis of V .

Suppose $v \in V$, then

$$\begin{aligned} T v &= T(\langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n) \\ &= s_1 \langle v, e_1 \rangle f_1 + \dots + s_n \langle v, e_n \rangle f_n. \end{aligned}$$

Remark. $T e_j = s_j f_j$ for each $j \in [1:n]$

$$\begin{aligned} [T]_{\mathcal{B}}^{\mathcal{B}'} &= \begin{bmatrix} s_1 & & 0 \\ & \ddots & \\ 0 & & s_n \end{bmatrix} \quad \text{when} \\ &\quad \mathcal{B} = \{e_1, \dots, e_n\}, \\ &\quad \mathcal{B}' = \{f_1, \dots, f_n\}. \end{aligned}$$

In other words every operator on V has a diagonal matrix with respect to some orthonormal bases of V provided that we are permitted to use two different bases rather than a single basis as customary when working with operators.

It is worthwhile to mention that SVD holds not only for linear operators on V , but also for linear transformations from V to W , i.e., for $T \in L(V, W)$. We present below a matrices version of the same without a proof.

Theorem (SVD for Matrices).

Let A be an $m \times n$ matrix with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ and $\sigma_{r+1} = \sigma_{r+2} = \dots = \sigma_n = 0$.

Then there exists two lists of orthonormal vectors $u_1, \dots, u_m \in F^m$ and $v_1, \dots, v_n \in F^n$ s.t.,

$$Av_i = \begin{cases} \sigma_i u_i & \text{for } i=1, 2, \dots, r, \\ 0 & \text{if } i=r+1, \dots, n, \end{cases}$$

or equivalently,

$$A = U \Sigma V^* = \sum_{i=1}^r \sigma_i u_i v_i^*,$$

where $U = \begin{bmatrix} | & & | \\ u_1 & \dots & u_m \\ | & & | \end{bmatrix}_{m \times m}$, $V = \begin{bmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{bmatrix}_{n \times n}$,

$$\Sigma = \begin{bmatrix} \underbrace{\begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r \end{bmatrix}}_{r \times r} & \underbrace{\begin{bmatrix} 0 & & 0 \\ & \ddots & \\ 0 & & 0 \end{bmatrix}}_{(n-r) \times (n-r)} \\ \underbrace{\begin{bmatrix} 0 & & 0 \\ & \ddots & \\ 0 & & 0 \end{bmatrix}}_{(m-r) \times (m-r)} & \underbrace{\begin{bmatrix} 0 & & 0 \\ & \ddots & \\ 0 & & 0 \end{bmatrix}}_{(m-r) \times (n-r)} \end{bmatrix}_{m \times n}$$

The singular values of a matrix A are the square roots of the eigenvalues of A^*A .

Example. We will find the singular value decomposition of $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ given the following information:

- Eigenvalues of $A^*A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ are

$\lambda_1 = 2$, $\lambda_2 = 1$ and $\lambda_3 = 0$, with corresponding eigenvectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$.

Normalizing, we get

$$\text{let } v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}.$$

v_1, v_2, v_3 is an orthonormal basis of $\mathbb{C}^3$.

The singular values of A are $\sigma_1 = \sqrt{2}$, $\sigma_2 = \sqrt{1} = 1$, $\sigma_3 = \sqrt{0} = 0$.
So $r = 2 = n$.

$$V = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix},$$

$$\Sigma = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \text{ To find } u_1, u_2 \text{ we compute}$$

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

$$u_2 = \frac{1}{\sigma_2} A v_2 = \frac{1}{1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

These vectors u_1, u_2 already form orthonormal basis of $\mathbb{R}^2$.

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus, SVD of A is

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

$$A = U \Sigma V^*.$$

Applications of SVD.

$$A = \sum_{i=1}^r \sigma_i u_i v_i^*$$

SVD has many applications in practice. Here we provide an illustration of its application in image compression, noting that any image can be represented as an $m \times n$ matrix with pixel values.

Original

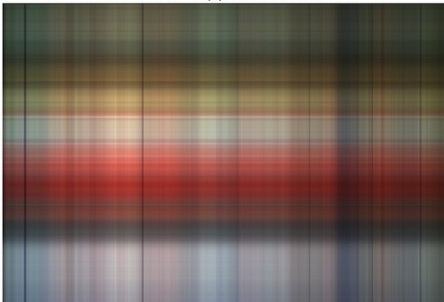


Pixel matrix $\rightarrow A_{1528 \times 1225}$

$$A = \sum_{i=1}^{\sigma} \sigma_i u_i v_i^*,$$

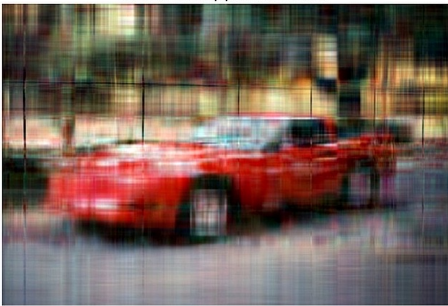
$\sigma \approx 1000$

Rank-1 Approximation



$$A_1 = \sigma_1 u_1 v_1^*$$

Rank-10 Approximation



$$A_{10} = \sum_{i=1}^{10} \sigma_i u_i v_i^*$$

Rank-100 Approximation



$$A_{100} = \sum_{i=1}^{100} \sigma_i u_i v_i^*$$

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^* \quad , 1 \leq k \leq r$$

is a low-rank approximation of

$$A = \sum_{i=1}^r \sigma_i u_i v_i^* .$$

In general, for most natural images, small k values, as little as $k=100$ (as illustrated in the above car image example) give a reasonable compression / approximation of the original image.